Star shredded by quasar 700 million years after Big Bang

JWST reveals a long-lived tidal disruption event in a dusty quasar at redshift 7.19.

arXiv:2609.30441 — Tidal disruption event in a z=7.19 quasar.

Authors: S. Fujimoto, Q. Fei, G. B. Brammer, K. Inayoshi, V. Kokorev, L. C. Ho, R. Li, F. Walter et al.

arXiv:2609.30441 doi:10.48550/arXiv.2609.30441

Background

GNz7q is a red quasar at z=7.19 hosting a 30-million-solar-mass black hole in a dusty starburst. Multi-epoch HST, Spitzer, and JWST imaging spanning two decades reveals a dramatic nuclear transient.

> **Tidal disruption events are rarely observed at cosmic dawn, leaving black-hole fueling in the reionization era largely unexplored.**

What this paper is selling

The transient faded by 1.1 magnitudes over two rest-frame years, with wavelength-dependent evolution and a t^-5/3 decline. A panchromatic model and a redshifted broad Balmer line point to the tidal disruption of a few-solar-mass star.

Key results to remember

- A 1.1-magnitude UV fade over two rest-frame years, far above quasar variability.
- Energetics match a tidal disruption event of a few-solar-mass star.
- A 1500 K dust echo at sub-parsec scales confirms circumnuclear reprocessing.

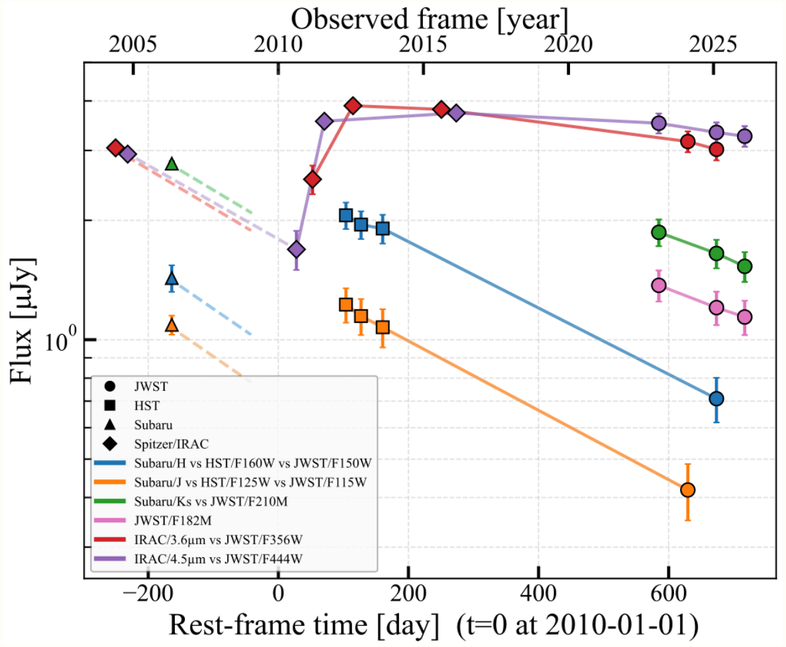


Figure 2 | Multi-epoch photometry shows smooth, wavelength-dependent dimming, the hallmark of a cooling transient rather than stochastic quasar variability.

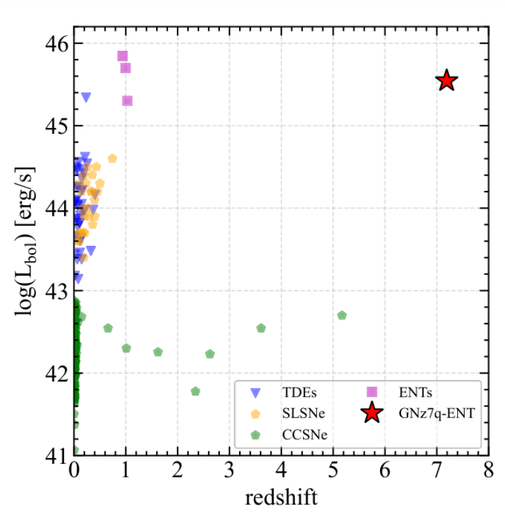


Figure 4 | GNz7q's peak luminosity and redshift place it among the most energetic known tidal disruption events, extending the population to the reionization era.